

Forecasting Government Spending on Corticosteroid Subsidies in Australia

Forecast and Conclusion

The final forecast indicates that government spending on corticosteroid subsidies in Australia is expected to continue rising over the next several months, with future seasonal peaks likely to exceed historical levels. This upward trend suggests increasing pressure on the national healthcare budget, particularly during winter seasons when demand for respiratory treatments is highest. Overall, the analysis concludes that spending will remain on a growth trajectory, reinforcing the need for proactive financial planning and timely policy adjustments to ensure uninterrupted access to essential medications.

Methodology

To generate a reliable forecast, several time-series models were evaluated, including Naive, Moving Average models (MA3, MA6, MA9), Simple Exponential Smoothing (SES), Holt–Winters, and ARIMA. While MA9 achieved the lowest statistical error values, it produced a flat future forecast, making it unsuitable for projecting seasonal peaks or long-term growth. Holt–Winters demonstrated the strongest balance between accuracy and practical forecasting value, as it effectively captured both the upward trend and pronounced seasonality observed in the data. This model was therefore selected as the final forecasting approach.

Data and Exploration Insights

The dataset contains monthly government spending on corticosteroid subsidies under the Pharmaceutical Benefits Scheme from 1991 to 2008, totaling 204 observations. Exploratory analysis revealed a clear upward trend across the 17-year period, with consistent seasonal spikes in winter months driven by higher demand for respiratory treatments. Spending growth accelerated after 2002, suggesting potential impacts from policy changes, increased prescription volume, or rising drug prices. These patterns highlight recurring financial pressures and underscore the importance of forecasting for budget planning.

Limitations and Recommendations

The primary limitation of the analysis lies in its reliance on historical patterns, which may not fully reflect future shifts in policy, drug pricing, population health trends, or unexpected events that can alter demand. The models also assume consistent seasonality, which may vary in real-world contexts. For future iterations, the analysis could be improved by incorporating external variables, such as pricing policies, demographic changes, inflation-adjusted costs, testing additional seasonal models, and comparing multiple forecast horizons. From a policy perspective, increasing budget allocation, reviewing reimbursement strategies, and closely monitoring monthly spending are recommended to prepare for rising subsidy requirements.